

CPS I — Exercise Sheet 3

1	2	3	Σ
4/4	3/3	4/8	11/15

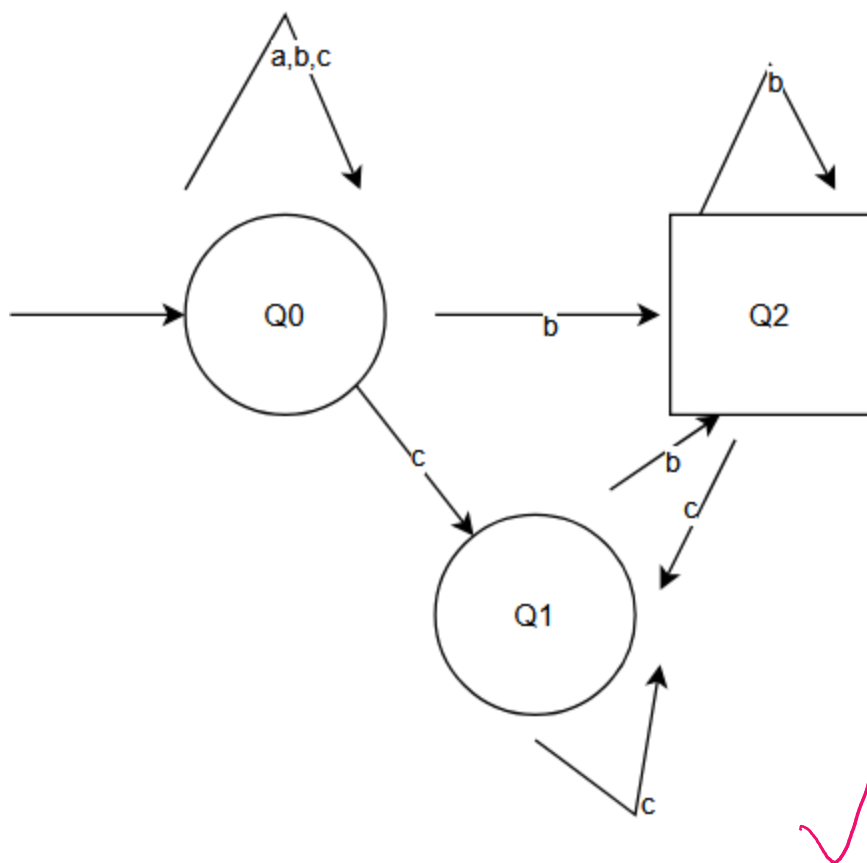
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1. Intersection of ω -regular languages

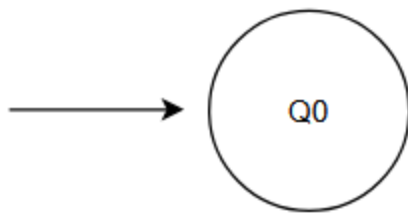
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(a)



(b)

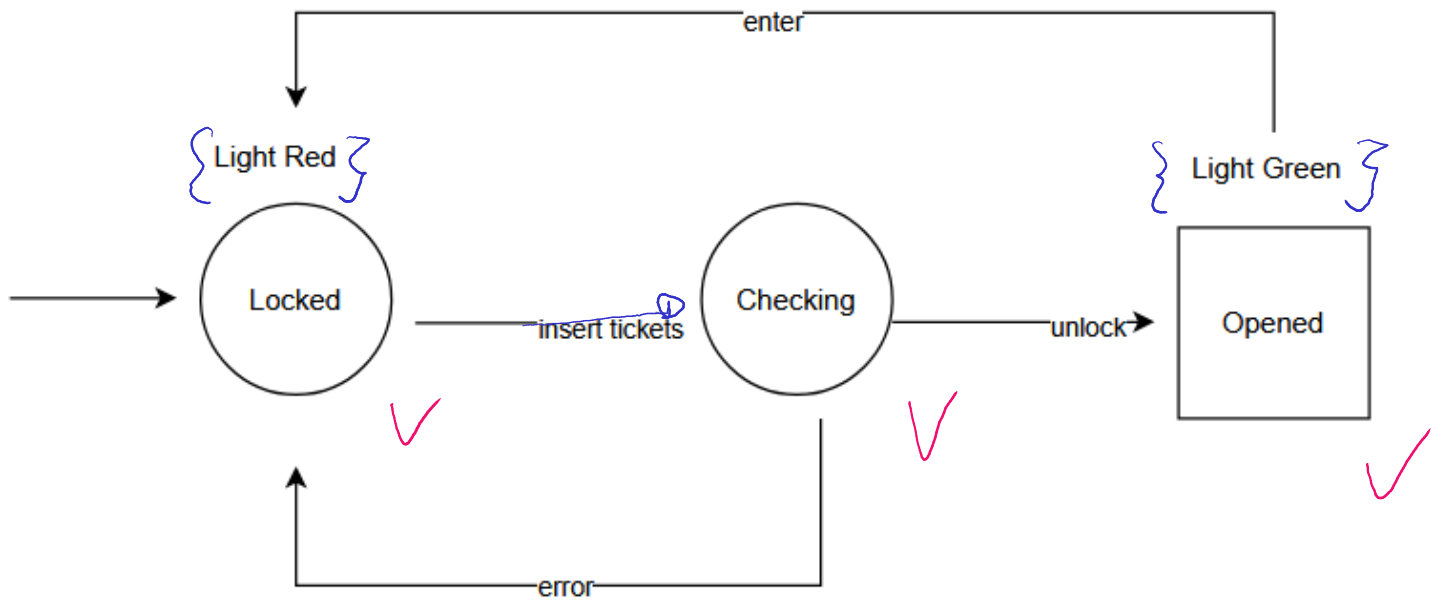
Empty ω Language



✓

2. Transition Systems

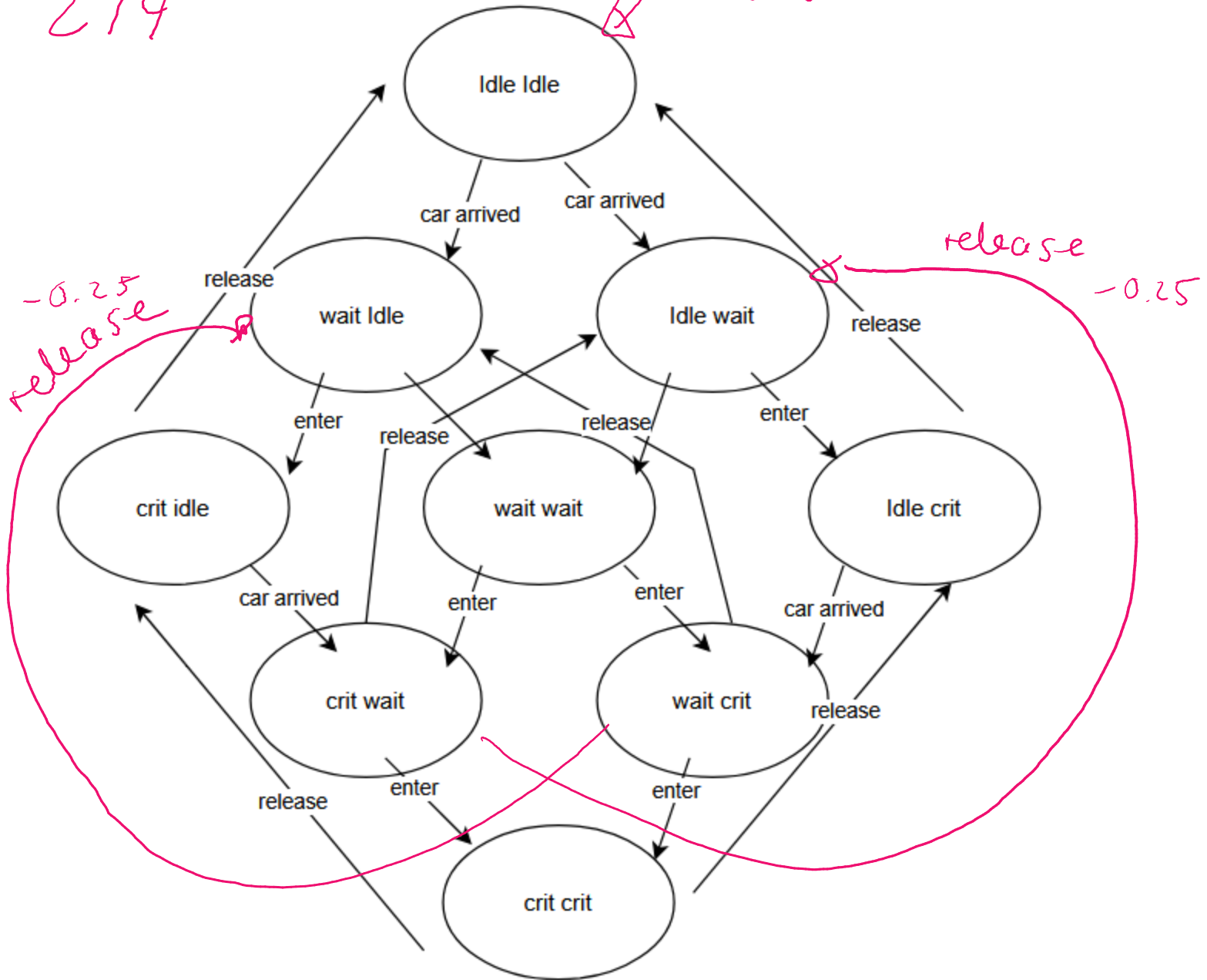
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This models an automated ticket gate

✓

3. Crossroads Traffic Lights



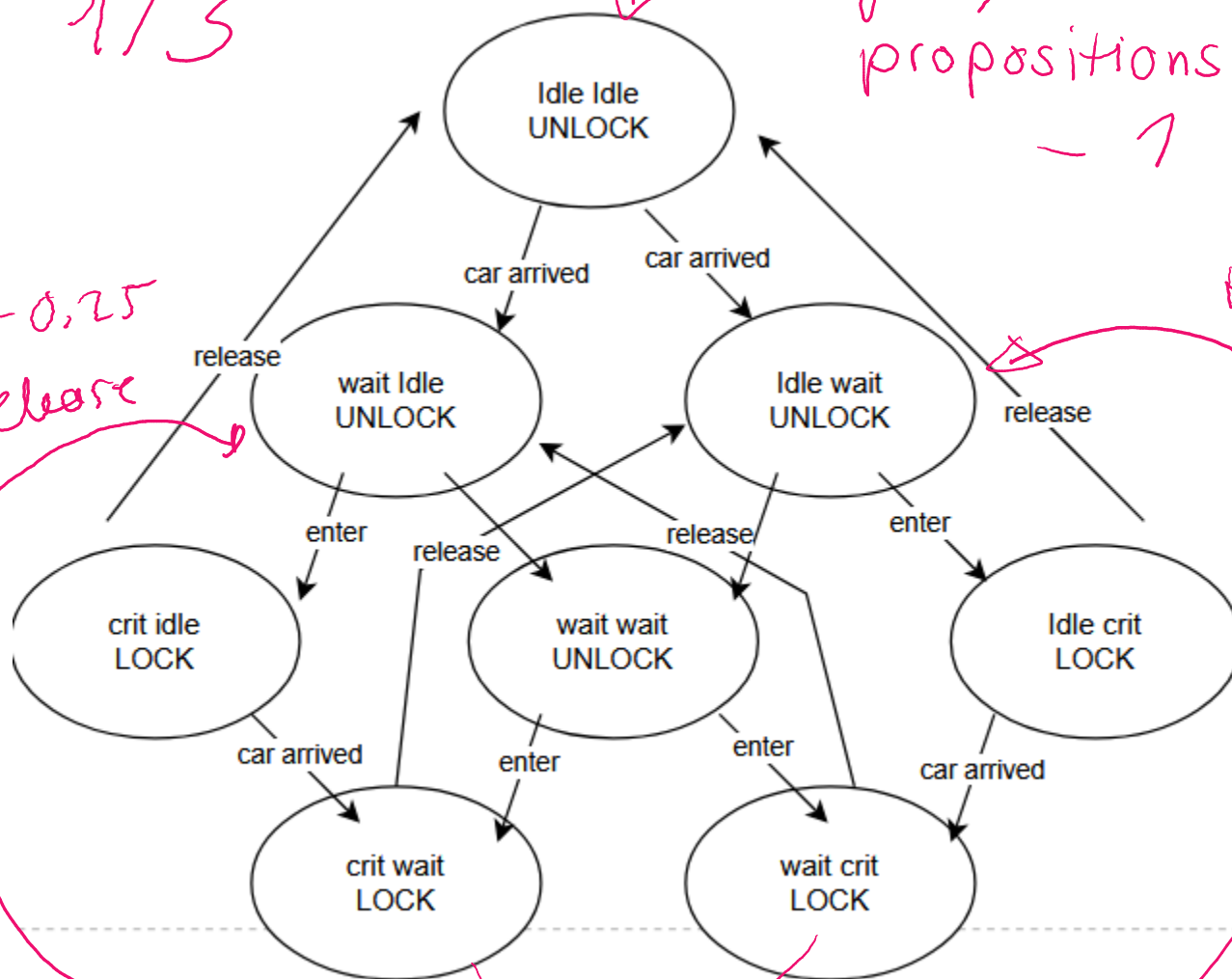
each state should be labeled with a set of atomic propositions: -1

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↓ -0.5 again, atomic propositions missing: - 1

-0.25 release

-0.25 release



hard to see, since you didn't include them--

- Yes, safe. The atomic propositions green1 and green2 correspond to crit_1 and crit_2, respectively. Once one direction acquires the lock, the other direction cannot execute unlock until the first one releases it. Therefore, there is no reachable state where both crit_1 and crit_2 hold simultaneously. ✓

- A common expectation is liveness / responsiveness / no starvation, i.e., if there is a car waiting in a direction (state wait_i), it will eventually be allowed to proceed (state crit_i). This property may not hold in the current model because the arbiter only enforces mutual exclusion but does not implement fair scheduling. ✓

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